strTrim [SimTalk]

Returns a string from which *Plant Simulation* removed leading and trailing blanks, tab stops, and carriage returns.

Type

Function

Syntax

```
strTrim(Text:string) → string
```

Parameter

The parameter *Text* of data type *string* designates the string, which contains leading and trailing blanks, tab stops, and carriage returns, i.e., whitespace characters.

Return Value

The return value has the data type *string*.

Examples

```
print strTrim(" ab c ") // returns "ab c"
```

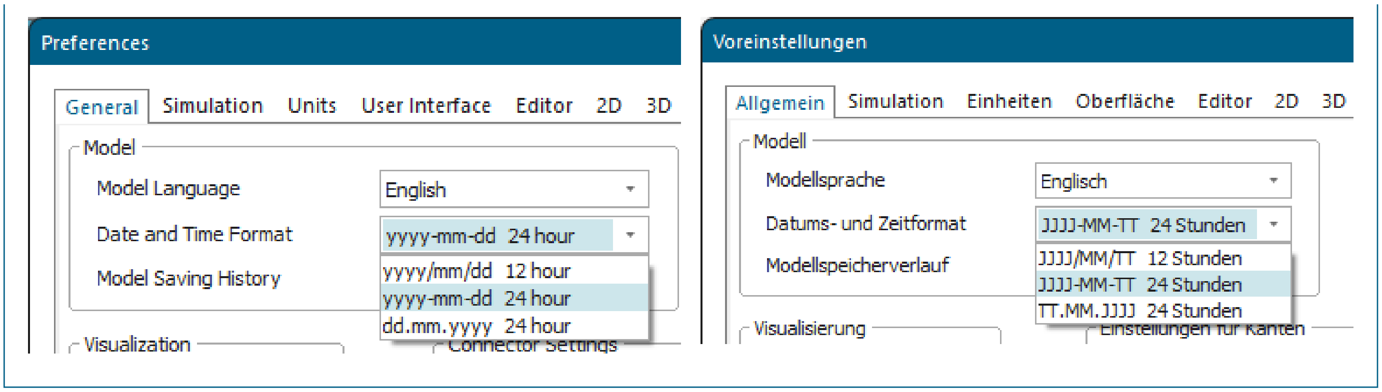
```
strTrim("  hello  ") = strTrim("hello    ") -- ignores leading and trailing
spaces
```

Functions for Managing Date and Time

Functions for Managing Date and Time

SimTalk provides the functions listed in the table of contents to the left for managing date and time information.

> **Note**
> *Plant Simulation* uses the **Date and Time Format** of the **Model Language** you selected.



See also

Date and Time Format [model settings]

Model Language [model settings]

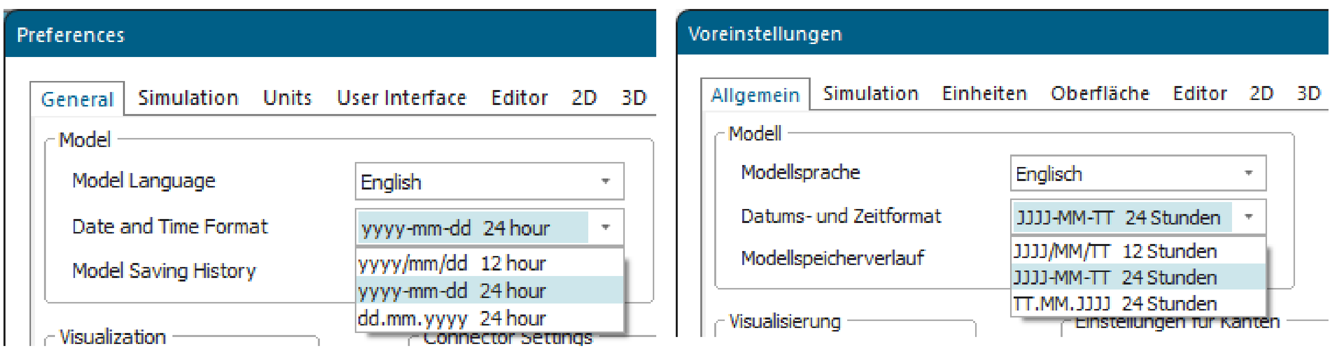
CalendarWeek [SimTalk]

Returns the calendar week of the specified date according to DIN 1355/ ISO 8601.

Remarks

A new week always starts on Monday.

According to DIN 1355/ ISO 8601 the first calendar week of a year is the first week that contains at least four days of the new year.

> **Note**
>
> *Plant Simulation* uses the **Date and Time Format** of the **Model Language** you selected.
>
> 
>

Type

Function

Syntax

```
CalendarWeek(Date:dateTime) → integer
```

Parameter

The parameter *Date* of data type *dateTime* designates the date.

Return Value

The return value has the data type *integer*.

Examples

```
print CalendarWeek(str_to_date("2023/11/21")) // returns 47, model language English
```

```
print CalendarWeek(str_to_date("21.11.2023")) // returns 47, model language German
```

See also

Date and Time Format [model settings]

Model Language [model settings]

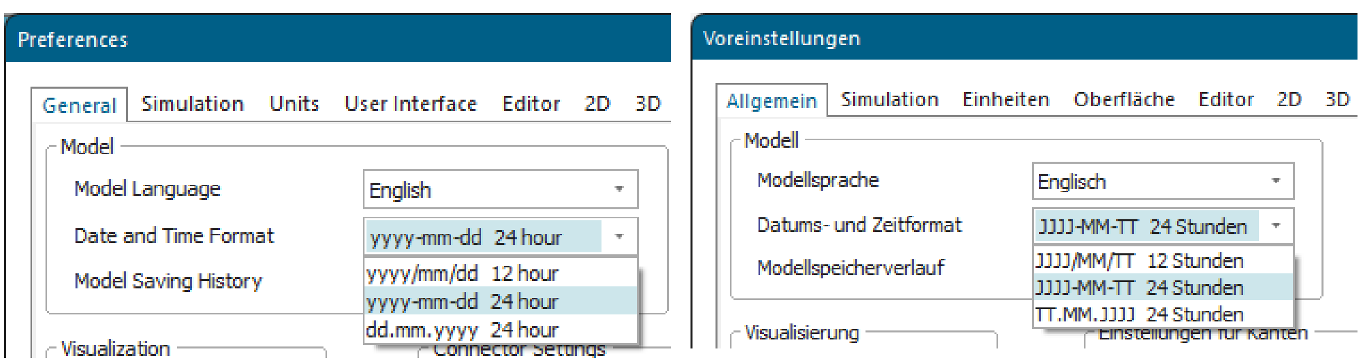
week [SimTalk]

CalendarYear [SimTalk]

Returns the current year of the **Gregorian Calendar**.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
CalendarYear(Value:dateTime) → integer
```

Parameter

The parameter *Value* of data type *dateTime* designates the date.

Return Value

The return value has the data type *integer*.

Example

```
print(CalendarYear(sysdate)) // 2024
print(year(sysdate))         // 124
```

See also

Date and Time Format [model settings]

Model Language [model settings]

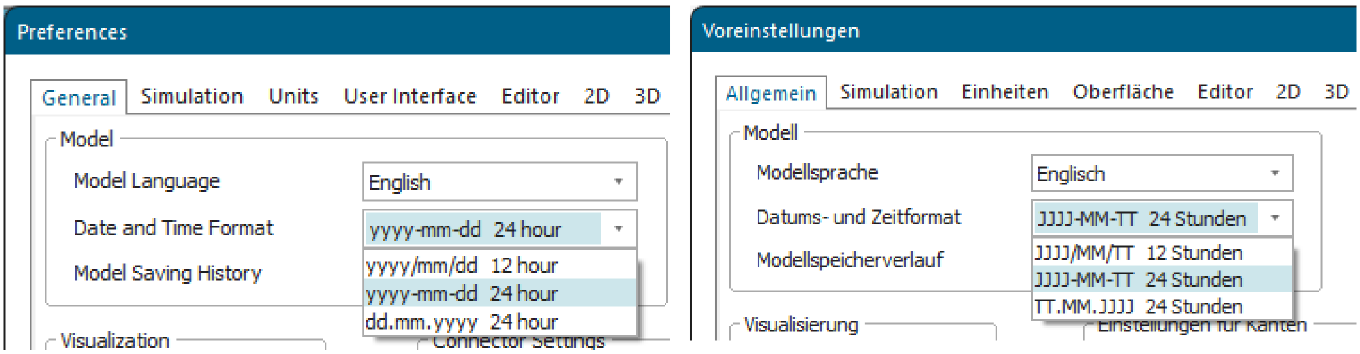
year [SimTalk]

day [SimTalk]

Extracts the day from a *dateTime* value.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
day(Date:date) → integer
day(DateTime:dateTime) → integer
```

Parameter

The parameter *Date* of data type *date* designates the date.

The parameter *DateTime* of data type *dateTime* designates the date and time.

Return Value

The return value has the data type *integer*.

Examples

```
print day(str_to_date("23/12/24"))           // returns 24, model language
English
print day(str_to_dateTime("24/1/1 1:00:00"))  // returns 1
```

```
print day(str_to_date("24.12.23"))           // returns 24, model language
German
print day(str_to_dateTime("1.1.24 1:00:00"))  // returns 1
```

See also

[Date and Time Format [model settings]]

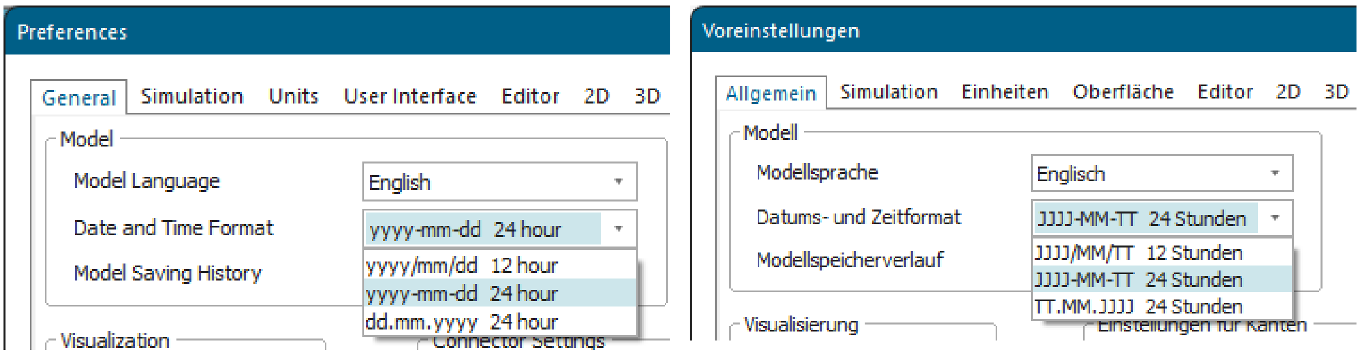
[Model Language [model settings]]

dayOfWeek [SimTalk]

Returns the number of days that has elapsed since the last Sunday.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
dayOfWeek(Date:date) → integer
dayOfWeek(DateTime:dateTime) → integer
```

Parameter

The parameter *Date* of data type *date* designates the date.

The parameter *DateTime* of data type *dateTime* designates the date and time.

Return Value

The return value has the data type *integer*.

Day of Week	Return Value
Monday	1
Tuesday	2
Wednesday	3

Day of Week	Return Value
Thursday	4
Friday	5
Saturday	6
Sunday	0

Examples

```
var d: date; var dt: dateTime
dt := str_to_dateTime("24/1/3 1:20:30")
d := str_to_date("24/1/1")
print dayOfWeek(dt)  // returns 3, model language English
print dayOfWeek(d)   // returns 1
```

```
var d: date; var dt: dateTime
dt := str_to_dateTime("3.1.24 1:20:30")
d := str_to_date("1.1.24")
print dayOfWeek(dt)  // returns 3, model language German
print dayOfWeek(d)   // returns 1
```

See also

Date and Time Format [model settings]

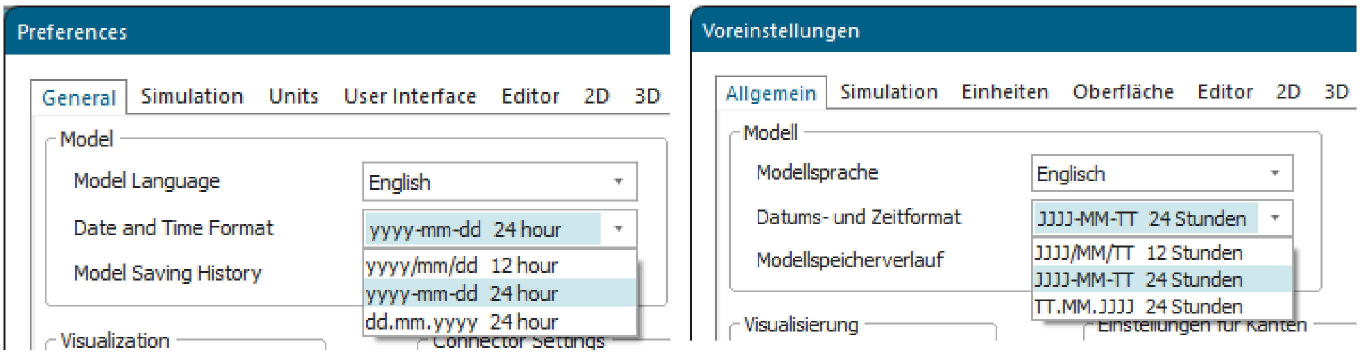
Model Language [model settings]

dayOfYear [SimTalk]

Returns the number of days that has elapsed since January first.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
dayOfYear(Date:date) → integer
dayOfYear(DateTime:dateTime) → integer
```

Parameter

The parameter *Date* of data type *date* designates the date.

The parameter *DateTime* of data type *dateTime* designates the date and time.

Return Value

The return value has the data type *integer*.

Examples

```
var d: date; var dt: dateTime
dt := str_to_dateTime("23/12/31 23:59:59")
d := str_to_date("24/1/1")
```

```
print dayOfYear(dt)  // returns 364, model language English
print dayOfYear(d)   // returns 0
```

```
var d: date; var dt: dateTime
dt := str_to_dateTime("31.12.23 23:59:59")
d := str_to_date("1.1.24")
print dayOfYear(dt)  // returns 364, model language German
print dayOfYear(d)   // returns 0
```

See also

Date and Time Format [model settings]

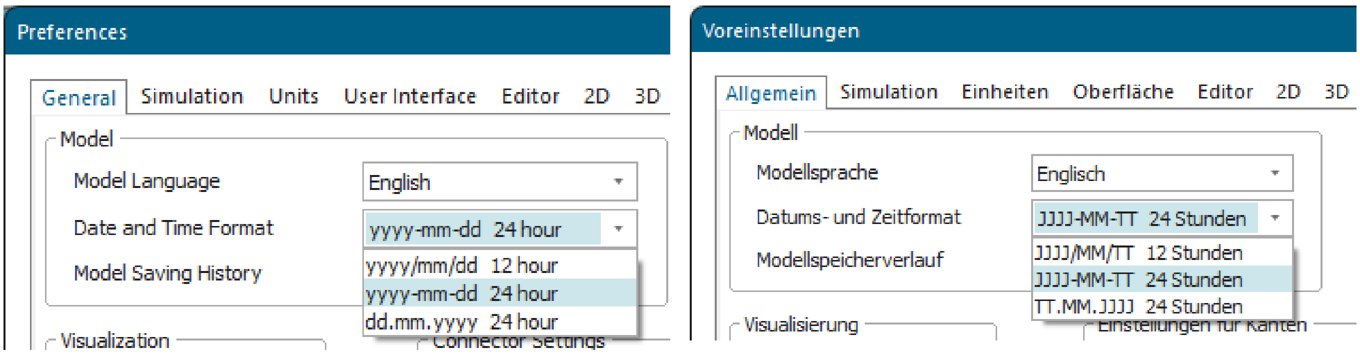
Model Language [model settings]

getDate [SimTalk]

Returns the date of a value containing date and time.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
getDate(Date/Datetime:dateTime)
```

Parameter

The parameter *Date* or *DateTime* of data type *date* or *dateTime* designates the value containing date and time.

Examples

```
var d: date; var dt: dateTime
dt := str_to_dateTime("2023/11/12 11:30:00")
d := getDate(dt)
print d // returns 2023/11/11, model language English
```

```
var d: date; var dt: dateTime
dt := str_to_dateTime("12.11.2023 11:30:00")
d := getDate(dt)
print d // returns 12.11.2023, model language German
```

See also

Date and Time Format [model settings]

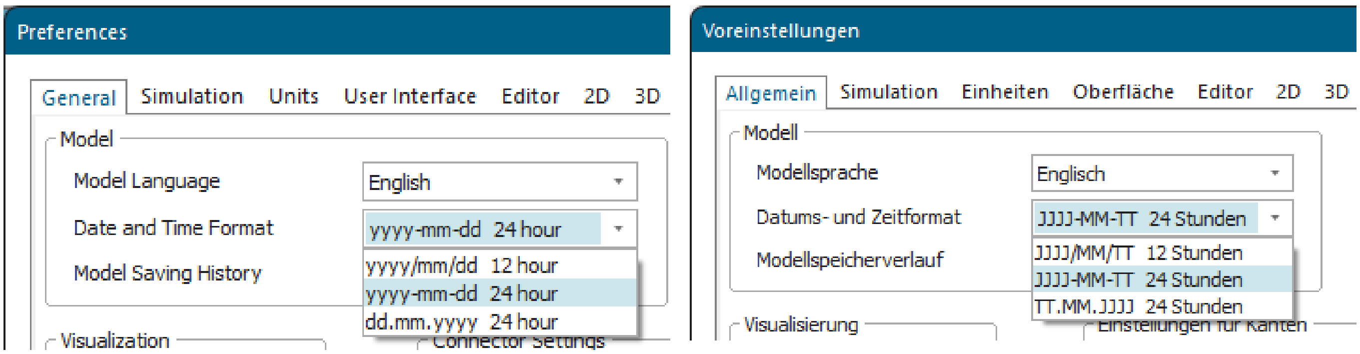
Model Language [model settings]

month [SimTalk]

Returns the month of a value containing date and time.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
month(Date:date) → integer
month(DateTime:dateTime) → integer
```

Parameter

The parameter *Date* of data type *date* designates the date.

The parameter *DateTime* of data type *dateTime* designates the date.

Return Value

The return value has the data type *integer*.

Example

```
print month(str_to_date("24.12.23"))          // returns 12
print month(str_to_dateTime("1.1.24 1:00:00"))   // returns 1
```

See also

Date and Time Format [model settings]

Model Language [model settings]

setDaylightSavingTime [SimTalk]

Sets the beginning and the end of **daylight saving time**.

Type

Function

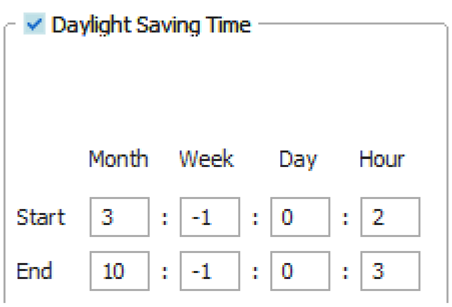
Syntax

```
setDaylightSavingTime(BeginningAndEndOfDaylightSavingTime:string)
```

Parameter

The parameter *BeginningAndEndOfDaylightSavingTime* of data type *string* consists of the following components.

The first set of four numbers sets the beginning of daylight saving time:



- monthOfYear
- weekOfMonth
- dayInWeek
- hour

The second set of four numbers sets the end of daylight saving time:

- monthOfYear

- weekOfMonth
- dayInWeek
- hour

To deactivate daylight saving time, specify an empty string `""`.

Example

```
setDaylightSavingTime("4,3,3,12,10,4,3,3")
setDaylightSavingTime("") // no daylight saving time
```

See also

Daylight Saving Time [preferences]

sysDate [SimTalk]

Returns the current system time of the computer on which *Plant Simulation* runs.

Title

Function

Syntax

```
sysDate → dateTime
```

Return Value

The return value has the data type *dateTime*.

The returned value has a resolution of 1 millisecond.

Example

```
print sysDate                             // current time
EventController.StartDate := sysDate    // apply to simulation
```

See also

getHighResolutionClock [SimTalk]

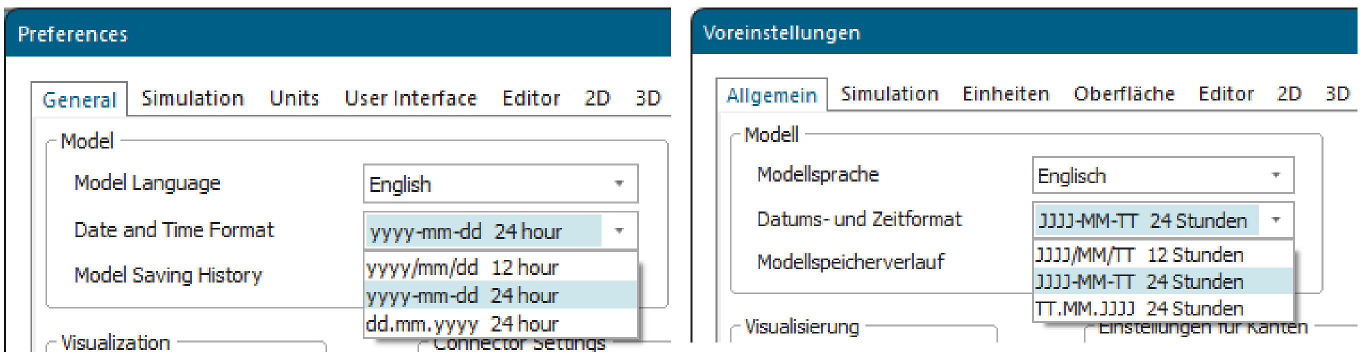
processTime [SimTalk]

timeOfDay [SimTalk]

Returns the time portion of a value containing date and time.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
timeOfDay(DateTime:dateTime) → time
```

Parameter

The parameter *DateTime* of data type *dateTime* designates the value containing date and time.

Return Value

The return value has the data type *time*.

Examples

```
var EclipseOfTheSun: datetime; var t: time; var d: date
EclipseOfTheSun := str_to_datetime("11.08.1999 11:30:00")
t := timeOfDay(EclipseOfTheSun)
print t // 11:30:00.0000
```

```
d := getDate(EclipseOfTheSun)
print d // returns 11.08.1999, model language English
```

```
var Sonnenfinsternis: datetime; var t: time; var d: date
Sonnenfinsternis := str_to_datetime("11.8.99 11:30:00")
t := timeOfDay(Sonnenfinsternis)
print t // 11:30:00.0000
d := getDate(Sonnenfinsternis)
print d // returns 11.08.1999, model language German
```

See also

Date and Time Format [model settings]

Model Language [model settings]

week [SimTalk]

Returns the number of weeks that started since January first of a year.

Remarks

The week always begins on a Monday. This week does not correspond to the calendar week.

Type

Function

Syntax

```
week(Date:date) → integer
week(DateTime:dateTime) → integer
```

Parameter

The parameter *Date* of data type *date* designates the date.

The parameter *DateTime* of data type *dateTime* designates the date and the time.

Return Value

The return value has the data type *integer*.

Example

```
print week(str_to_date("1.1.24"))             // returns 1
print week(str_to_date("1.2.24"))             // returns 5
print week(str_to_dateTime("1.1.24 1:00:00")) // returns 1
```

See also

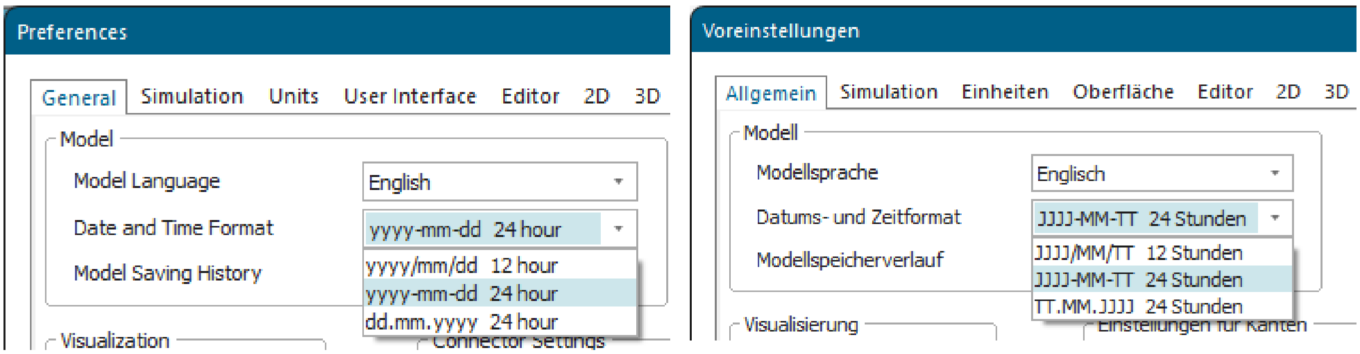
CalendarWeek [SimTalk]

year [SimTalk]

Computes the number of years elapsed since the year 1900.

Remarks

Plant Simulation uses the **Date and Time Format** of the **Model Language** you selected.



Type

Function

Syntax

```
year(Date:date) → integer
year(DateTime:dateTime) → integer
```

Parameter

The parameter *Date* of data type *date* designates the date.

The parameter *DateTime* of data type *dateTime* designates the date and the time.

Return Value

The return value has the data type *integer*.